

Beyond Forecast Accuracy: Feasibility-Constrained Demand Planning

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Abstract

Firms increasingly use predictive demand models to set inventory targets, replenishment plans, and execution commitments. However, operations execute planning signals rather than forecast accuracy metrics. A WAPE-improving model can still create plan movements that exceed the organization's ability to adapt. This paper proposes a forecast-to-plan feasibility gate for demand-planning deployment. The gate converts candidate forecasts into planning signals, measures planning-execution gap rate, and separates diagnostic forecast accuracy from operational planning utility. Across 123 paper-facing evaluation groups, the WAPE-best deployable strategy differs from the operational-loss-best deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. The results show that ensemble, smoothing, and DP-based policies occupy different deployment regimes: ensembles can dampen model-specific swings, smoothing slows adaptation when execution capacity is tight, and DP or budgeted DP supports governance when switching decisions have future consequences. The empirical contribution is not only that forecast accuracy is incomplete, but that the planning-execution gap can be measured and used as a deployment gate for forecast-driven planning strategies. WAPE should remain diagnostic; it should not be the sole deployment approval criterion.

Keywords

Demand planning, Forecast-to-plan conversion, Operational feasibility, Planning stability, Execution risk.

1 Introduction

Demand planning is an operational deployment problem, not only a forecasting problem. Forecast-driven decisions influence inventory targets, replenishment orders, supplier coordination, warehouse labor, transportation commitments, and planning-system governance. A model rollout that improves WAPE can still create abrupt plan changes, new approval requirements, or execution workloads that the operating system cannot absorb.

Operations teams execute planning signals, not forecast error metrics. A planner cannot execute WAPE or RMSE; the organization executes an inventory target, a staffing plan, a shipment commitment, or a replenishment quantity. When the planning signal changes faster than supplier, warehouse, transportation, or enterprise-system workflows can adapt, the forecast layer and execution layer become misaligned. This paper calls that misalignment the planning-execution gap.

Forecasting teams optimize error metrics, but operations teams absorb plan changes. Recent work shows rapid progress in AI-enabled supply-chain planning and forecasting benchmarks (Walter et al. 2025; Culot et al. 2024; Vlachos and Reddy 2025; Makridakis et al. 2022). These advances make deployment evaluation more important. More accurate forecasts can be valuable, but forecast accuracy alone does not show whether the resulting plans are stable, feasible, and governable.

This paper studies feasibility-constrained model selection for operational demand planning. Candidate forecasts and planning policies are converted into executable planning signals, evaluated by a forecast-to-plan feasibility gate, and ranked through deployable decision-layer strategies. The empirical contribution is not only that forecast accuracy is incomplete, but that the planning-execution gap can be measured and used as a deployment gate for forecast-driven planning strategies.

The study uses public data in distinct operational roles. DataCo supply-chain execution records anchor execution-risk scenarios from observed late-delivery heterogeneity. Favorita daily retail demand data provide the main forecast-to-plan proof of concept. M5 retail data test hierarchy and intermittent-demand robustness. Walmart weekly sales data

provide secondary business-context and cadence robustness. Rossmann is reserved for future work because completed Rossmann experiment outputs were not found in the audit.

The paper makes four IEOM-oriented contributions:

1. Define planning-execution gap rate as a measurable deployment-feasibility metric.
2. Propose a forecast-to-plan feasibility gate that separates WAPE diagnostics from operational planning utility.
3. Compare deployable strategy families, including ensemble, smoothing, greedy, DP, and budgeted-DP policies.
4. Validate that deployable ranking changes across execution-risk scenarios, planning hierarchy, intermittency, and verified Walmart business-context and cadence settings.

1.1 Objectives

The specific objectives of this study are:

1. Define a forecast-to-plan feasibility gate that measures operational feasibility in addition to WAPE.
2. Measure the planning-execution gap using execution violation rate and planning-execution gap rate.
3. Construct DataCo-informed execution-risk scenarios for deployment stress testing.
4. Compare accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable planning strategies.
5. Validate ranking shifts across Favorita, M5, and Walmart, and translate the findings into practical planning recommendations.

2 Literature Review

2.1 AI-enabled Demand Planning and Forecasting Benchmarks

Recent supply-chain research shows increasing interest in AI-enabled demand planning. Walter et al. (2025) review artificial intelligence in demand planning and identify AI tools, supply-chain applications, and digital supply-chain impacts as major knowledge clusters. Culot et al. (2024) review empirical AI studies in supply-chain management and emphasize the need to connect AI adoption to operational research directions. Vlachos and Reddy (2025) similarly review machine learning in supply-chain management and identify planning, execution, and control as areas where machine learning can support decision making.

Forecasting benchmarks and time-series foundation models show the same trend from a forecasting perspective. The M5 competition demonstrates the scale and hierarchy of modern retail forecasting benchmarks (Makridakis et al. 2022). Recent time-series foundation models, including decoder-only forecasting models and pretrained probabilistic time-series models, show that forecasting pipelines are moving toward broader zero-shot and cross-domain capability (Das et al. 2024; Ansari et al. 2024). These studies motivate the forecasting layer of the present paper, but they do not remove the need to evaluate whether forecast-driven planning signals are operationally executable.

2.2 Forecast-to-decision and Predict-then-optimize Research

Forecast-to-decision research is directly relevant because it explains why prediction accuracy and downstream decision quality can differ. Elmachtoub and Grigas (2022) show that the standard predict-then-optimize workflow can be improved by training predictions toward decision loss rather than prediction loss alone. Mandi et al. (2024) survey decision-focused learning and describe how machine learning and constrained optimization can be integrated to

improve decision quality under uncertainty. Jia et al. (2025) propose a scenario predict-then-optimize method for data-driven online inventory routing, showing that downstream routing and replenishment decisions require more than point forecast accuracy.

The present paper is related to this stream, but its purpose is different. It does not introduce a new end-to-end learning algorithm. It proposes an interpretable forecast-to-plan decision layer for operational demand planning, where existing candidate forecasts are evaluated by inventory exposure, planning volatility, switching burden, and execution-capacity violations.

2.3 Inventory Implications and Planning Stability

Inventory theory provides the operational foundation for holding and shortage costs. Silver et al. (1998) and Axsater (2015) describe inventory planning as a cost and service-level problem rather than a pure prediction problem. Forecast accuracy metrics remain important, but Hyndman and Koehler (2006) show that forecast-error measures have known limitations and must be interpreted carefully across series and contexts. Forecast combination is also relevant because weighted or averaged forecasts can reduce dependence on a single model (Bates and Granger 1969).

Modern forecasting candidate sets often include nonlinear machine-learning methods such as random forests, XG-Boost, and LightGBM (Breiman 2001; Chen and Guestrin 2016; Ke et al. 2017). These models can improve forecast accuracy, but model deployment also creates governance, monitoring, and switching burden. From an operations perspective, Forrester (1961) shows that industrial systems can amplify instability when material and information flows respond with delays. Advanced planning systems similarly separate forecasting, planning, coordination, and execution layers (Stadtler et al. 2015). This separation motivates explicit measurement of planning volatility and execution-capacity violations.

Rolling-horizon decision theory also supports finite-horizon selection logic because a locally attractive action can be undesirable once future transitions are considered (Sethi and Sorger 1991). This perspective is consistent with the paper's use of greedy-to-DP comparisons to measure myopic planning suboptimality.

2.4 Research Gap

Existing studies improve forecasts, combine forecasts, or optimize downstream decisions. Fewer studies provide a practical decision layer that compares forecasting-driven strategies by inventory exposure, planning volatility, switching burden, and execution-capacity violations. This paper addresses that gap by defining planning-execution gap rate and feasibility-adjusted normalized planning loss. The goal is not to replace forecasting metrics, but to position WAPE as a diagnostic measure alongside operational feasibility.

3 Methods

The proposed method is a deployment gate for selecting demand-planning strategies. Figure 1 shows the gate: candidate forecasts and policies are converted into planning signals, scored for feasibility, and then approved or rejected for deployment. This design follows forecast-to-decision research showing that downstream operational quality can differ from forecast-error quality (Elmachtoub and Grigas 2022; Mandi et al. 2024).

Figure 1 is a deployment gate, not only a workflow. WAPE remains visible as a diagnostic signal, but the approval decision is governed by whether the forecast-driven plan can be operationally executed.

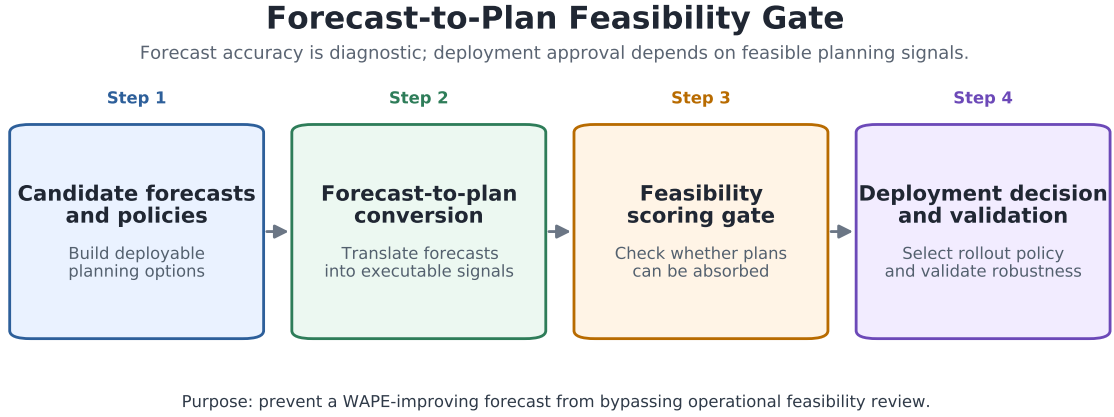


Figure 1: Forecast-to-plan feasibility gate for demand-planning deployment.

3.1 Forecast-to-plan Feasibility Model

Let i index a planning unit, t index a planning period, and m index a candidate forecasting model or planning policy. The candidate forecast is $\hat{y}_{i,t}^m$, and the executable planning signal is $p_{i,t}^m$. The forecast-to-plan conversion is:

$$p_{i,t}^m = g(\hat{y}_{i,t}^m, \text{policy}_t). \quad (1)$$

In this study, the planning signal is an inventory target implied by the candidate forecast and planning policy.

Inventory exposure is measured using holding and shortage terms:

$$IC_{i,t}^m = h[p_{i,t}^m - y_{i,t}]^+ + b[y_{i,t} - p_{i,t}^m]^+, \quad (2)$$

where h is holding cost, b is shortage cost, $y_{i,t}$ is realized demand, and $[x]^+ = \max(x, 0)$.

Planning volatility and execution violation are:

$$PV_{i,t}^m = |p_{i,t}^m - p_{i,t-1}|, \quad (3)$$

$$EV_{i,t}^m = [|p_{i,t}^m - p_{i,t-1}| - C_{i,t}]^+, \quad (4)$$

where $C_{i,t}$ is the execution capacity available to absorb plan movement. The planning-execution gap rate for strategy s is:

$$PEG(s) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{EV_t(s) > 0\}. \quad (5)$$

The normalized planning loss is:

$$L(s) = \beta_{\text{inventory}} \cdot \text{NIC}(s) + \lambda_{\text{volatility}} \cdot \text{NVC}(s) + \lambda_{\text{execution}} \cdot \text{NEC}(s) + \lambda_{\text{switch}} \cdot \text{NSC}(s), \quad (6)$$

where NIC, NVC, NEC, and NSC are normalized inventory, volatility, execution, and switching components. WAPE is reported separately as diagnostic forecast accuracy. The main operational objective excludes direct forecast-error terms because realized inventory exposure and feasibility terms capture the deployable consequences of forecast-driven planning decisions.

3.2 Stepwise Feasibility-constrained Selection Algorithm

Algorithm 1. Feasibility-constrained selection procedure.

1. Generate candidate forecasts or planning policies for each planning unit and period.
2. Convert each candidate forecast into a planning signal.
3. Compute inventory exposure relative to realized demand.
4. Compute planning volatility and switching burden relative to the previous executed plan.
5. Compute execution violation under the available execution capacity.
6. Aggregate the normalized planning loss.
7. Select the deployable planning action using one of the decision policies.
8. Validate the selected strategy against scenario and oracle benchmarks.

Steps 1–3 create candidate planning signals. Steps 4–6 score those signals by feasibility. Step 7 selects a deployment action. Step 8 validates whether the action remains preferred under execution-risk scenarios and oracle benchmarks. The procedure is modular: accuracy-first baselines, ensembles, smoothing rules, greedy selectors, DP selectors, and budgeted-DP selectors all pass through the same forecast-to-plan feasibility gate.

3.3 Horizon-aware and Constrained Selection

The greedy selector minimizes the current-period operational cost. Let $c_t(m_{t-1}, m)$ denote the validation-derived stage cost of choosing candidate m after previous model m_{t-1} . The greedy selector chooses:

$$m_t^G = \arg \min_{m \in \mathcal{M}} c_t(m_{t-1}, m). \quad (7)$$

The finite-horizon DP selector accounts for future transition effects:

$$V_t(m_{t-1}) = \min_{m \in \mathcal{M}} \{c_t(m_{t-1}, m) + V_{t+1}(m)\}, \quad (8)$$

with terminal condition $V_{T+1}(\cdot) = 0$. The greedy-to-DP gap measures myopic selection suboptimality. When this gap is small, one-step selection may be adequate; when switching approval and coordination costs are material, DP provides a planning-ahead benchmark.

The budgeted DP selector adds a governance constraint:

$$\sum_{t=2}^T \mathbb{I}\{m_t \neq m_{t-1}\} \leq K. \quad (9)$$

The state includes the previous model and the number of switches already used. Transitions exceeding the switch budget are discarded. If missing candidate availability leaves no feasible path, the implementation uses a flagged incumbent-stays fallback that is budget-compliant when the incumbent model remains available.

Smoothing represents gradual operational adaptation:

$$p_t = \alpha p_t^{\text{candidate}} + (1 - \alpha) p_{t-1}. \quad (10)$$

Lower α values reduce execution burden but can delay response to demand changes. Ensemble policies combine candidate forecasts and are evaluated as low-disruption alternatives when accuracy remains comparable.

3.4 Validation Benchmarks

The Realized-Inventory Oracle DP is non-deployable. It uses realized test-period inventory outcomes over the same forecast candidates and measures the value of unavailable inventory-outcome information. The DP-oracle gap measures selection suboptimality relative to this non-deployable benchmark.

The Perfect-Demand Oracle is also non-deployable. It uses realized demand as the planning signal and is a perfect-demand diagnostic reference, not a guaranteed lower bound. Exact demand tracking can increase planning volatility and execution violations when the operation cannot absorb rapid plan movement. The perfect-demand reference gap therefore measures a diagnostic total gap that includes both forecast uncertainty and feasibility effects.

4 Data Collection and Experimental Design

The empirical design uses DataCo, Favorita, M5, and Walmart for distinct operational roles. Table 1 summarizes these roles. DataCo is not treated as an inventory-control dataset; it transfers empirical heterogeneity in execution risk into scenario design. Favorita is the main forecast-to-plan proof-of-concept dataset. M5 tests hierarchy and intermittent-demand robustness. Walmart is included as verified secondary business-context robustness.

Table 1: Dataset roles and experimental purpose.

Dataset	Role	Data used	Purpose
DataCo	Execution-risk anchor	Late-delivery outcomes by operational context	Defines data-informed execution-risk scenarios; not treated as an inventory-control dataset.
Favorita	Main proof of concept	Daily store-family retail demand panel	Tests forecast-to-plan conversion and deployable strategy ranking under execution-risk scenarios.
M5	Robustness analysis	Hierarchical and intermittent retail demand	Tests whether the planning-execution gap persists across planning grain and demand sparsity.
Walmart	Business-context robustness	Weekly store-department sales with holiday and markdown context	Tests context-aware forecasting, holiday and markdown windows, weekly cadence constraints, and switch-budget sensitivity.

DataCo execution-risk anchor. DataCo contains supply-chain fulfillment records with late-delivery outcomes. The analysis extracts late-delivery anchors from context-level percentiles. The current generated anchors range from 0.535 at the low-risk percentile to 0.901 at the severe-risk percentile. Under the additive clipped scenario mapping, these anchors produce execution weights from 0.318 to 0.500. These are sensitivity weights, not exact dollar costs for Favorita, M5, or Walmart.

Favorita proof of concept. Favorita is a daily retail demand-planning panel from Corporacion Favorita. The pipeline treats each store-family pair as a planning unit and uses promotion, holiday, oil-price, transaction, and store-context information where available at planning time. Candidate forecasts include transparent baselines and machine-learning candidates such as LightGBM and XGBoost when dependencies are available (Ke et al. 2017; Chen and Guestrin 2016). The purpose is to test whether forecast-to-plan conversion changes deployable strategy ranking.

M5 robustness design. M5 contains 3,049 items across 10 stores, and the pipeline samples item-store series for quick and medium robustness runs (Makridakis et al. 2022). The IEOM submission uses M5 to evaluate item-store, department-store, and category-store planning grains and to test intermittent-demand buckets. This design checks whether the planning-execution gap persists when demand is sparse or aggregated.

Walmart business-context robustness design. Walmart provides weekly store-department sales with holiday, markdown, store, and economic context fields. The IEOM submission uses the generated Walmart outputs to compare history-only and history-plus-context candidate forecasts, holiday and markdown stress windows, weekly cadence

constraints, and switch-budget sensitivity. This design checks whether the feasibility layer remains informative when the planning problem is weekly and context driven rather than daily and store-family driven.

Rossmann is not treated as a completed experiment in this IEOM draft. It remains an optional future robustness panel.

5 Results and Discussion

5.1 Numerical Results

Finding 1. WAPE-best is often not deployment-best. Across 123 paper-facing evaluation groups, the WAPE-best deployable strategy differs from the operational-loss-best deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. These verified values show that the forecast-to-plan feasibility gate measures a deployment problem that WAPE alone does not capture. Table 2 summarizes the core numerical evidence.

Table 2: Core numerical evidence for the planning-execution gap.

Metric	Verified value	Interpretation
Accuracy-first mismatch rate	91.1% of 123 groups	WAPE-best deployable strategy often differs from operational-loss-best deployable strategy.
Mean accuracy-first loss gap	11.169	Average normalized-loss disadvantage of accuracy-first selection when it is not operationally best.
Accuracy-first PEG rate	15.4%	Fraction of periods with positive execution violation for accuracy-first strategies.
Best deployable PEG rate	2.7%	Same metric for the best deployable operational-loss strategy.
Greedy-to-DP gap	0.044	Measures myopic selection cost where greedy and DP are both available.
Mean DP-oracle gap	0.198	Selection suboptimality relative to non-deployable realized-inventory DP information.
DataCo execution-risk anchors	0.535 to 0.901	Late-delivery anchors used for low to severe execution-risk scenarios.

Finding 2. Ensemble is a low-disruption first rollout candidate. Favorita quick-mode results provide the main proof of concept. In the baseline scenario, Global Best has WAPE 0.109 and normalized planning loss 1.550, while Operational Ensemble has WAPE 0.119 and normalized planning loss 1.422. Under the severe DataCo-informed execution-risk scenario, Global Best remains accurate but has normalized planning loss 2.000. Simple Ensemble and Operational Ensemble have lower normalized losses of 1.473 and 1.496, respectively. Ensemble averaging dampens model-specific forecast swings while retaining adaptation, which makes it a practical low-disruption candidate when execution pressure makes plan movement costly.

Finding 3. Smoothing and DP serve different governance regimes. Smoothing reduces execution burden by slowing adaptation, which is useful when execution capacity is tight. DP reduces myopic switching by planning over the horizon, which is useful when switching approval and coordination matter. The mean greedy-to-DP gap is 0.044, and the mean DP-oracle gap is 0.198. These values do not imply that DP is always required; they show where planning-ahead value becomes material relative to a one-period selector. M5 item-store results also illustrate the stability boundary: Smoothed Alpha 0.25 is the best deployable method at the item-store grain, with WAPE 0.368, PEG rate 0.010, and normalized planning loss 1.223.

Finding 4. Deployment policy depends on operating context. M5 reveals that planning aggregation changes strategy optimality: Operational Ensemble is strongest at department-store and category-store grains, while Smoothed Alpha 0.25 is strongest at the item-store grain and across low-, medium-, and high-intermittency buckets. Low-level item-store planning is sparse and volatile, so smoothing can dampen noise; department and category aggregation stabilize demand, allowing ensemble methods to perform better. Walmart adds secondary business-context validation: the

main Walmart robustness summary contributes 20 evaluation groups with an 80.0% accuracy-first mismatch rate, and switch-budget sensitivity shows that larger switch budgets improve WAPE slightly while increasing normalized loss. The deployment implication is that one global model-approval rule should not be applied uniformly across planning grain, demand sparsity, cadence, and business-context regimes.

5.2 Graphical Results

Figure 2 plots the Favorita accuracy versus execution-penalty frontier. The figure shows that WAPE and execution penalty do not move together. Global Best is near the best WAPE region, but other deployable strategies produce much lower execution penalties. The operational implication is that the deployable frontier is not necessarily the WAPE-best point.

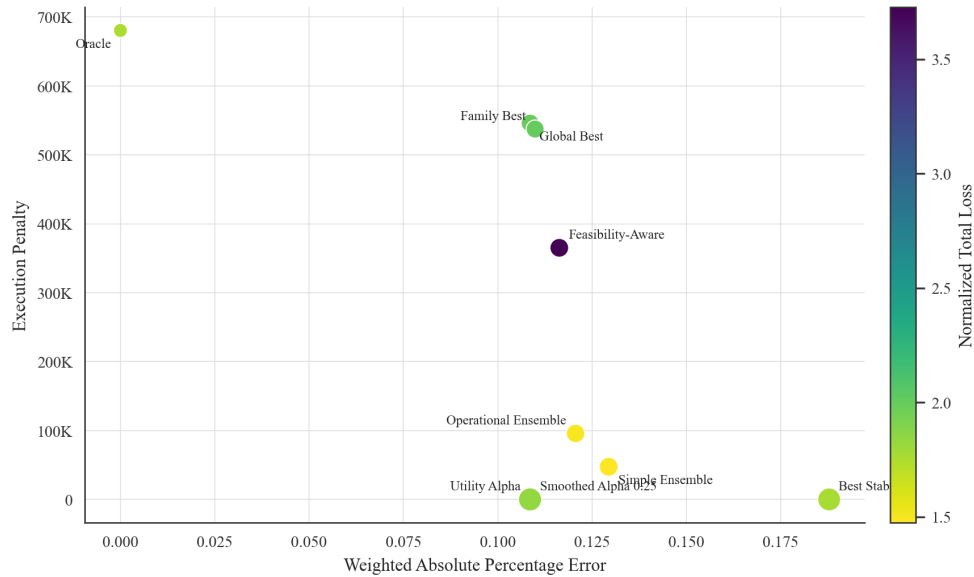


Figure 2: Favorita accuracy and execution penalty frontier.

Figure 3 shows normalized planning loss across DataCo-informed execution scenarios. As execution-risk weight changes, strategy ranking changes. This supports scenario testing as a deployment practice: the model selected for a flexible execution environment may not be the model selected for a constrained execution environment.

5.3 Proposed Model-Rollout Protocol

The empirical results translate into a five-step model-rollout protocol. First, screen candidate models by WAPE to remove clearly weak forecasts. Second, convert surviving forecasts into planning signals. Third, measure PEG rate and normalized planning loss. Fourth, stress-test the candidate under execution-risk scenarios. Fifth, select the deployment strategy by operating condition rather than by WAPE alone. Table 3 summarizes the recommended policy choices.

The protocol is intentionally not a single-method prescription. Flexible operations may approve an accuracy-first model or ensemble after feasibility screening. Constrained operations should consider ensemble or smoothing policies. High-governance environments should evaluate DP or budgeted DP because the timing and number of switches become part of the deployment decision.

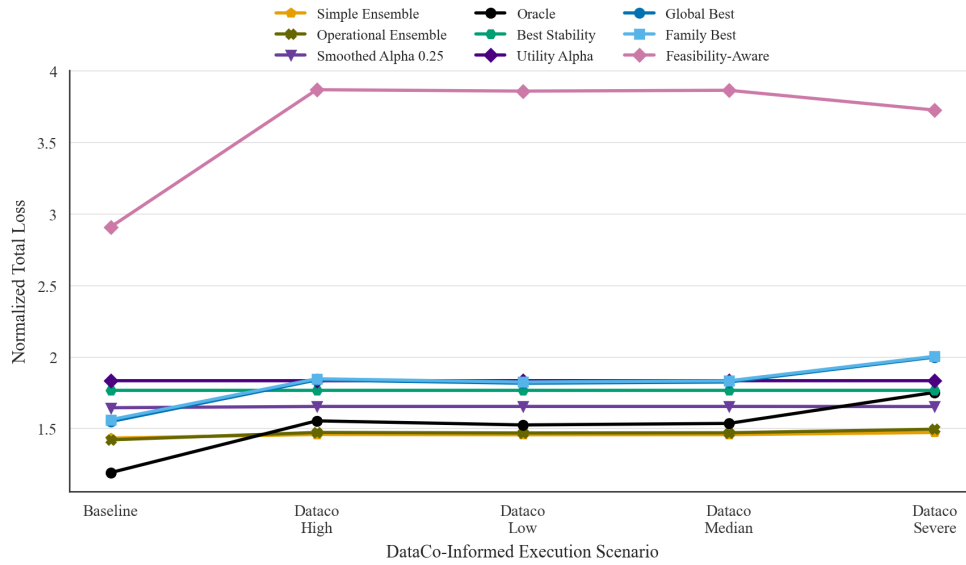


Figure 3: Favorita normalized planning loss by DataCo-informed execution scenario.

Table 3: Deployment strategy families and rollout conditions.

Operating condition	Recommended deployment policy	Reason
Flexible execution capacity	Accuracy-first or ensemble	WAPE gains can be useful when plan movement is absorbable, but feasibility screening remains required.
Moderate execution constraint	Ensemble	Averaging can dampen model-specific swings while retaining adaptation.
Severe execution constraint	Smoothing	Gradual adaptation can reduce execution burden when abrupt plan movement is costly.
High switching governance	DP or budgeted DP	Horizon-aware selection accounts for future switching and coordination burden.
Multi-level hierarchy	Choose by planning grain	Item, department, and category planning can favor different deployment mechanisms.

5.4 Validation

5.4.1 Baseline Validation

The mismatch rate and PEG-rate contrast show that WAPE-best selection and deployment-best selection answer different questions. The study does not claim that feasibility-aware strategies always dominate; it shows that the operational deployment criterion is measurably different from the forecast-accuracy criterion.

5.4.2 Scenario Validation

DataCo-informed scenarios validate whether execution pressure changes deployable ranking. The Favorita baseline-to-severe shift shows that a strategy attractive under lower execution pressure can become less attractive when execution-capacity violations are weighted more heavily.

5.4.3 Algorithmic Validation

The greedy-to-DP gap measures the cost of one-period selection. The DP-oracle gap measures selection suboptimality relative to a non-deployable realized-inventory DP benchmark over the same forecast candidates. These gaps show how much value comes from horizon-aware selection and how much value remains unavailable without realized inventory-outcome information.

The perfect-demand diagnostic reference serves a different purpose. It is not a lower-bound oracle because exact demand tracking may create planning volatility that violates execution capacity. It therefore measures a total reference gap that includes forecast uncertainty and feasibility effects.

5.4.4 Robustness Validation

Favorita provides the main forecast-to-plan proof of concept. M5 shows that hierarchy and intermittency change the preferred deployable method. Walmart shows that weekly cadence and business context also change deployable strategy comparisons. Together, these results support the thesis that feasibility-constrained planning is a tradeoff surface, not a single-method leaderboard.

6 Conclusion

This paper reframes operational demand planning as feasibility-constrained model selection through a forecast-to-plan feasibility gate. The planning-execution gap was quantified by comparing accuracy-first model selection with operational-loss-optimal deployable planning selection. Across 123 paper-facing evaluation groups, the accuracy-first deployable method differs from the operational-loss-best deployable method in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, while the best deployable operational-loss strategy has a gap rate of 2.7%.

Objective 1 was addressed by defining a forecast-to-plan feasibility gate that reports WAPE separately from operational planning utility. Objective 2 was addressed by defining planning-execution gap rate. Objective 3 was addressed by using DataCo execution records to construct data-informed execution-risk scenarios. Objective 4 was addressed by comparing accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable strategies. Objective 5 was addressed through Favorita, M5, and Walmart validation and through evidence-linked planning recommendations.

The results show that execution-risk scenarios can change deployable strategy ranking, that hierarchy and intermittency can change the preferred mechanism, and that weekly business context can affect feasibility tradeoffs. The practical implication is direct: planners should evaluate deployment feasibility before model rollout. Forecast accuracy remains important, but it should be interpreted as a diagnostic metric rather than as the sole deployment approval criterion.

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Acknowledgement placeholder. Replace this paragraph with funding, advisor, or institutional acknowledgements before submission if applicable.

Biography

Author biography placeholder. Replace this paragraph with a biography for each author before submission. IEOM guidance limits each biography to 250 words.

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